

Food Rescue AI

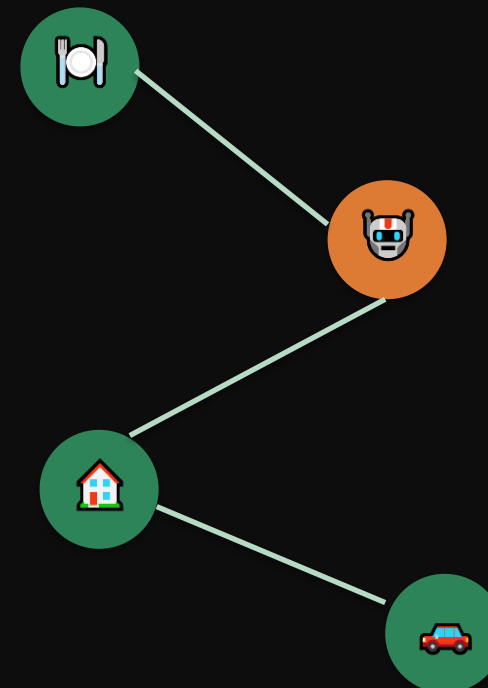
Don't Waste Food. Rescue It.

Surplus-to-Shelter — Real-Time Food Rescue Routing

A working full-stack prototype for real-time food rescue.

SKIT College

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Problem Statement

02

Edible surplus often has only a few hours to be rescued.

The challenge is coordinating the right shelter, driver and timing before expiry.



Surplus food

Restaurants may have safe food that cannot be sold in time.



Shelter matching

Shelters need the right food type and enough available capacity.



Last-mile coordination

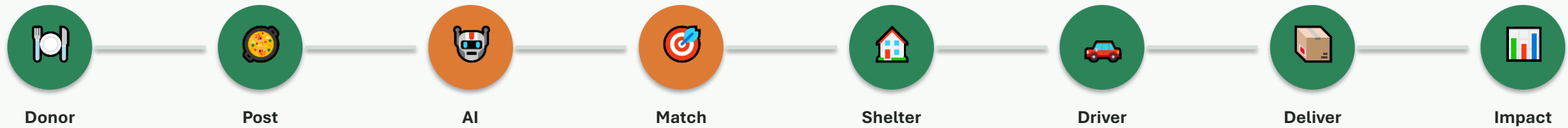
Drivers must reach pickup and delivery locations on time.

Goal: connect surplus food with the right shelter and driver while the food is still usable.

Proposed Solution

03

Food Rescue AI turns a food listing into a complete rescue operation.



Automated matching

Scores shelters using distance, capacity, compatibility, urgency and driver availability.



Physical verification

GPS checks the driver at pickup and shelter arrival.



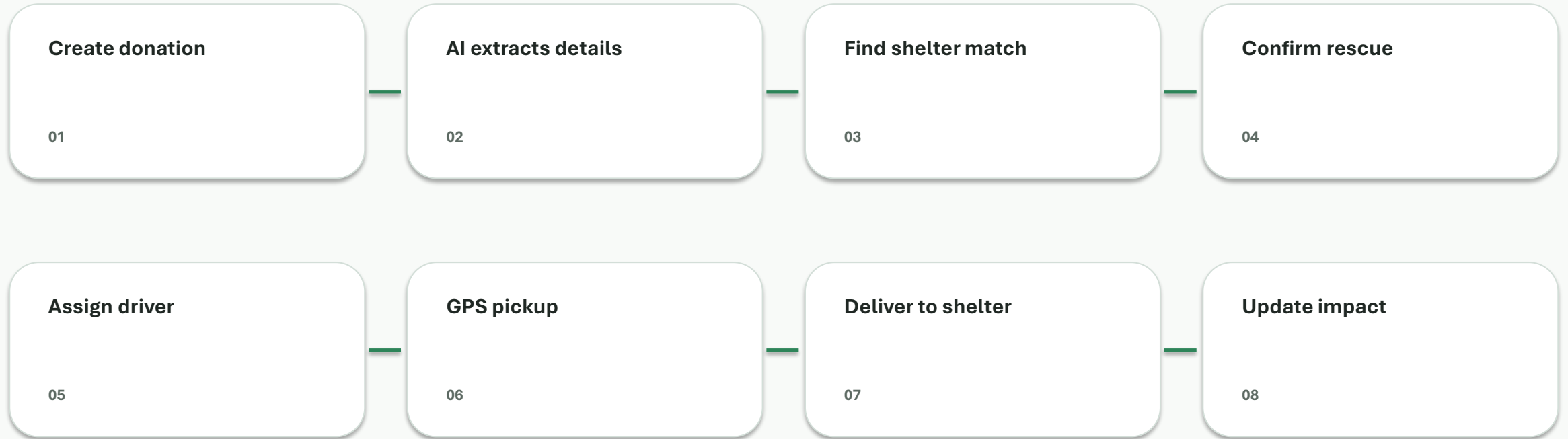
Closed-loop impact

Successful delivery updates rescue and impact records.

One platform. One rescue workflow. Less food waste.

End-to-End Workflow

04



Rescue status



The rescue moves through a defined status lifecycle from match to delivery.

AI + Smart Matching

05



AI food extraction

Input: "15 kg cooked rice and vegetables available for the next 2 hours."

Food Type → Cooked Rice & Vegetables

Quantity → 15 kg

Category → Cooked Meal

Urgency → High



Expiry awareness

Urgency increases as expiry approaches; expired donations are blocked from matching.

35% distance + 25% capacity + 20% compatibility + 15% urgency + 5% driver availability

Shelter match score

Distance



35
%

Capacity



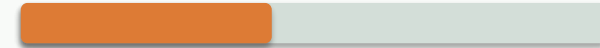
25
%

Food compatibility



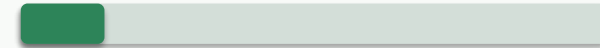
20
%

Urgency



15
%

Driver availability



5%



Decision

Suitable shelters are ranked by the weighted score before rescue confirmation.

Key Features

06



Food Donation

Quantity + expiry +
location



AI Extraction

Food details from text



Smart Matching

Weighted shelter ranking



Expiry Tracking

Urgency-aware matching



Driver Coordination

Accept and update rescue



GPS Verification

500 m pickup/delivery
check



Map Integration

Leaflet + OpenStreetMap



Impact Dashboard

Food, meals and CO₂
estimates

Technology + Architecture

07

Stack

Frontend
React • Vite • Tailwind • Router

Backend
Node.js • Express • JWT

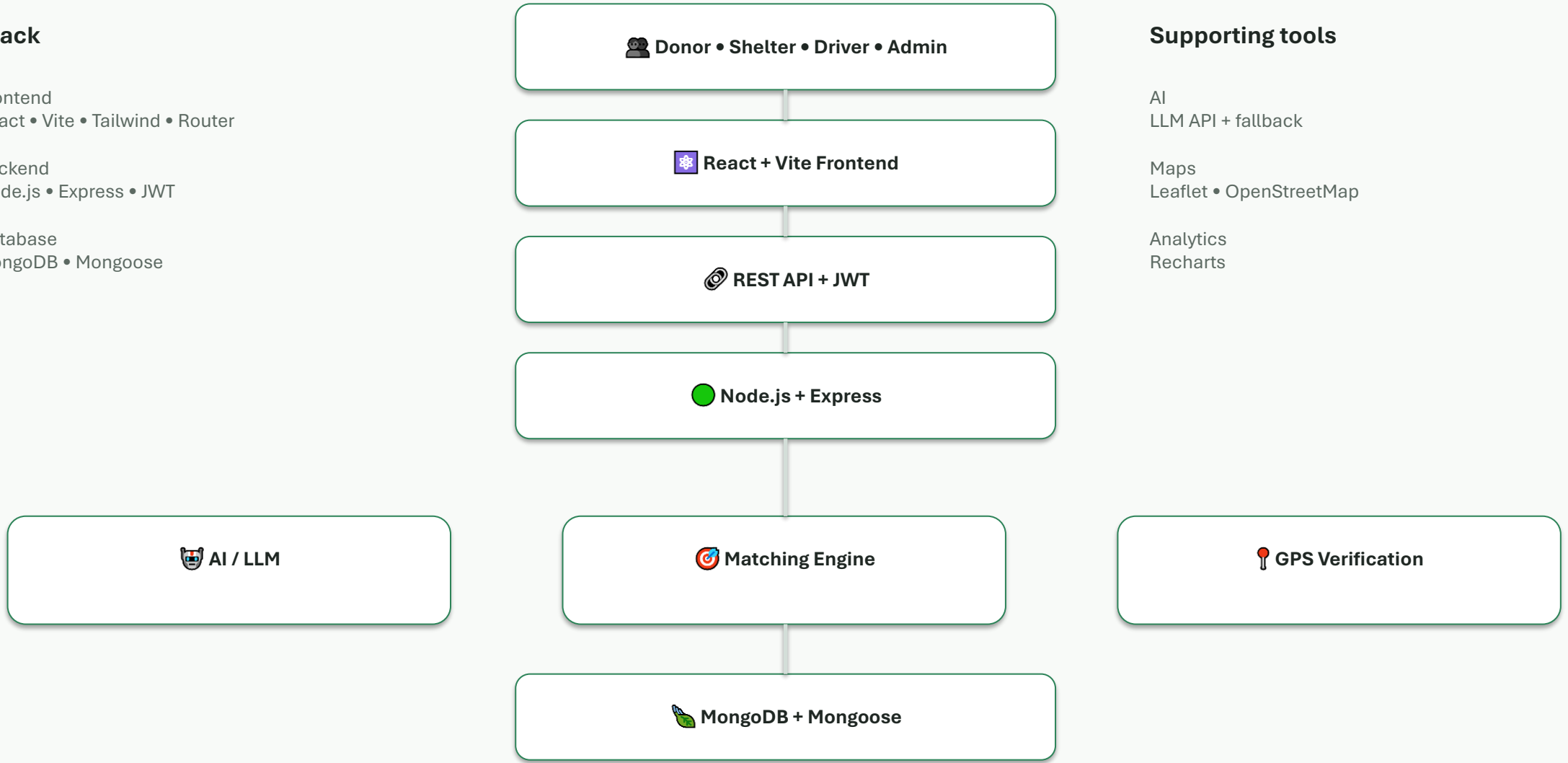
Database
MongoDB • Mongoose

Supporting tools

AI
LLM API + fallback

Maps
Leaflet • OpenStreetMap

Analytics
Recharts





FoodRescue AI

Don't Waste Food. Rescue It.



Real-World Impact

- Reduces food waste
- Helps people in need
- Supports sustainable cities
- Measurable social and environmental impact

Users (Frontend)



Food Donor

Posts surplus food donation (food type, quantity, location, expiry time)



Shelter / NGO

Manages capacity, accepted food types and receives donations



Volunteer Driver

Accepts rescue assignments, picks up and delivers food



Admin

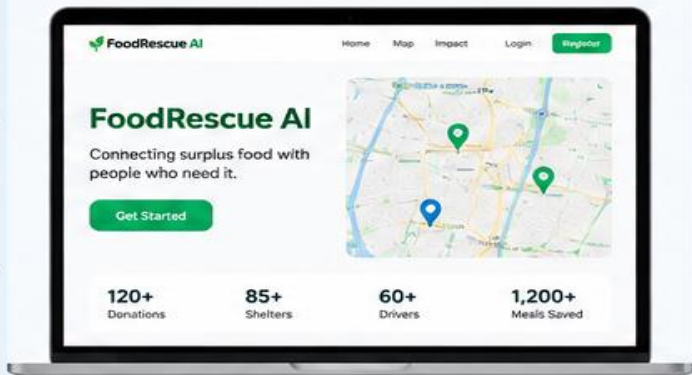
Monitors platform and impact metrics

Frontend (Client)



React + Vite

Tailwind CSS + React Router
Framer Motion + Leaflet + Recharts



HTTP Requests
(REST API)

JSON Response

Backend (Server)



Node.js + Express

REST APIs
Authentication (JWT)
Business Logic
Matching & GPS Verification

Auth APIs

Donation APIs

Shelter APIs

Driver APIs

Matching APIs

Rescue Status APIs

Impact APIs

Database



MongoDB + Mongoose

- Users (Donors, Shelters, Drivers)
- Donations (Surplus Food)
- Shelters (Capacity & Requirements)
- Drivers (Availability & Location)
- Rescues (Status Tracking)
- Impact Data (Food Rescued, Meals, CO₂ Avoided)



LLM / AI Service

Food Information Extraction

- Extracts food type, category, quantity, unit and expiry time from natural language description
- Supports OpenAI API with demo fallback mode

Smart Matching Engine



35% Distance (Geographical proximity)

25% Shelter Capacity (Available space)

20% Food Compatibility (Food type match)

15% Urgency (Expiry time)

5% Driver Availability (Available drivers)

End-to-End Food Rescue Workflow



1. Post Donation

Donor posts surplus food



2. AI Extraction

Extract food information using LLM



3. Smart Matching

Find best suitable shelter



4. Driver Assignment

Assign available driver



5. GPS Pickup Verification

Verify driver arrival at pickup location



6. GPS Delivery Verification

Verify driver arrival at shelter



7. Impact Update

Update rescued food, meals and CO₂ impact

External Services & APIs



Leaflet + OpenStreetMap

- Interactive maps
- Location search
- Route visualization



Browser Geolocation API

- Get driver's current location
- Verify pickup and delivery
- 500-meter verification radius



LLM API (OpenAI)

- Natural language processing
- Structured food data extraction
- Demo fallback when API key is not provided



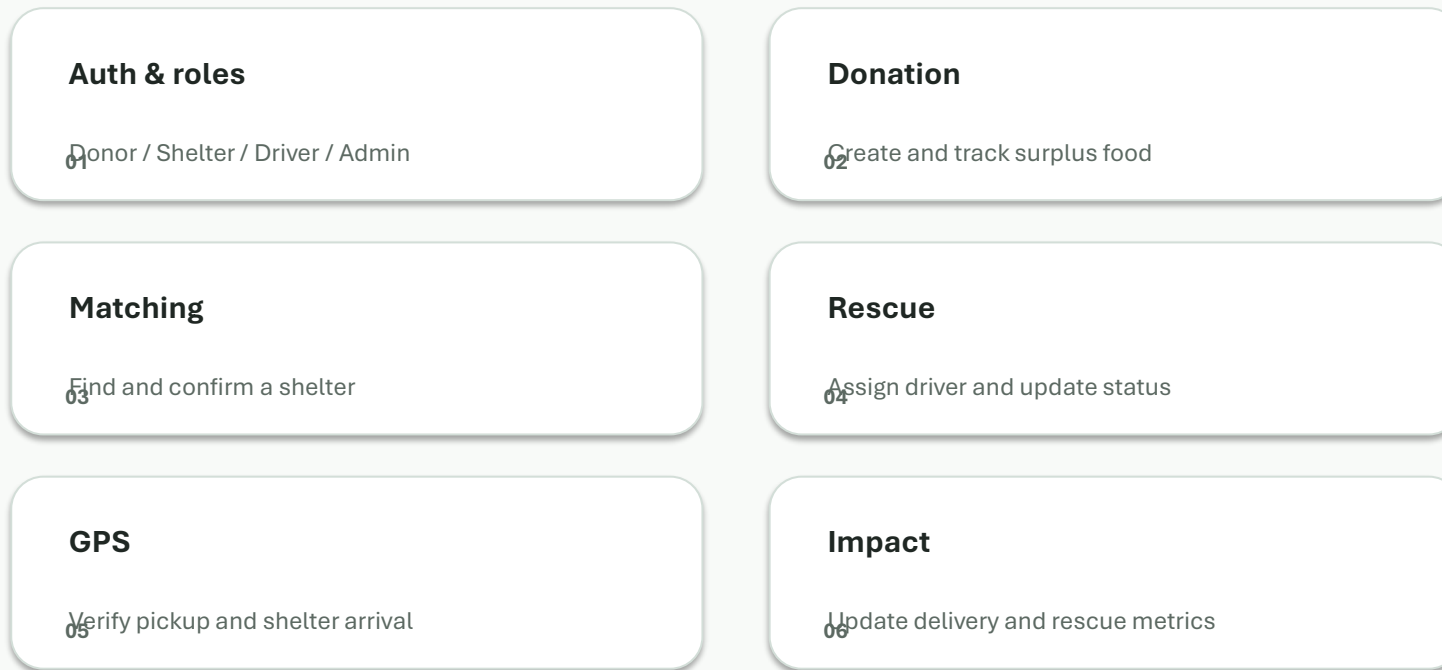
Impact Dashboard

- Total food rescued
- Successful deliveries
- Estimated meals saved
- Estimated CO₂ avoided
- Donation and rescue analytics

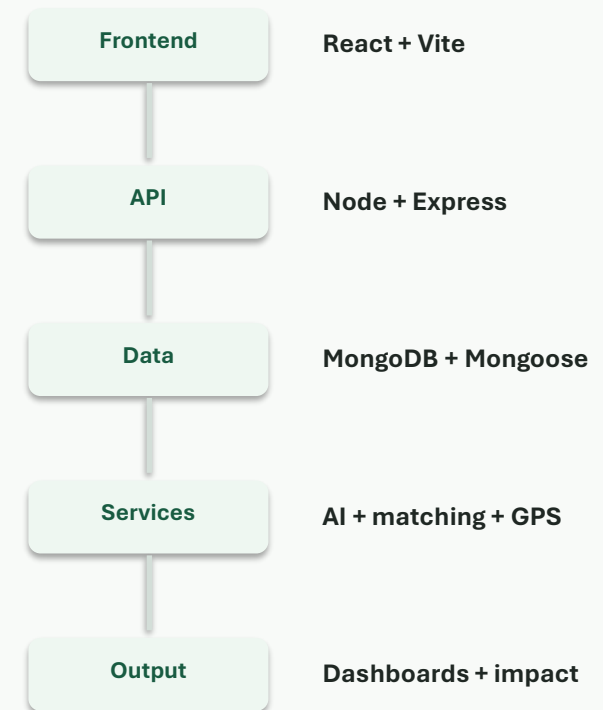
Build a Prototype

08

The prototype is a working end-to-end application, not a static mock-up.



Prototype path



Working demo • real API • real DB

Live Demo + Impact

Demo scenario



Green Leaf Restaurant

15 kg cooked rice & vegetables • 2 hour expiry



Hope Shelter

Suitable shelter selected by matching



Rahul Sharma

Assigned driver for the rescue

Example rescue impact

15 kg

Food rescued

42

Meals saved

37.5 kg

CO₂e avoided

1

Successful deliveries

Live journey

Create donation

AI extraction

Shelter match

Driver assigned

GPS pickup

Delivery

Impact update

ADD REAL APP SCREENSHOTS HERE

Donor Dashboard • Matching • Driver Dashboard • Impact

Use actual screenshots — no fabricated UI.

FUTURE SCOPE

From a working prototype to a city-scale rescue network.

 Mobile application

 Real-time driver tracking

 Route optimization

 Real-time notifications

 Better AI matching

 Predictive food demand

 Multi-city expansion

 NGO & government partnerships

“Surplus food should reach
people, not landfills.”



Food Rescue -AI

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Thank You • Questions?